

ENME608 Mechanisms

Velocity Analysis (Norton Ch. 6)

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Velocity Analysis — Overview

Velocity analysis follows directly from position analysis: *differentiate the vector loop equation once with respect to time*. The link lengths are constant, so only the angles change, and $\frac{d}{dt}(Re^{j\theta}) = jR\dot{\theta}e^{j\theta} = jR\omega e^{j\theta}$. The factor j rotates each velocity vector 90° ahead of its position vector.

Prerequisite: you must already have θ_3 and θ_4 from position analysis (same assembly mode — open or crossed).

Open-book: Lec 4 (velocity), Tutorial 5.

Exam focus. Default to the **analytical method**. The instant-centre and relative-velocity (graphical) methods are useful sanity checks but the closed-form equations are examinable.

Four-Bar Angular Velocities

Differentiating the four-bar loop $ae^{j\theta_2} + be^{j\theta_3} - ce^{j\theta_4} - d = 0$:

$$a\omega_2 e^{j\theta_2} + b\omega_3 e^{j\theta_3} - c\omega_4 e^{j\theta_4} = 0.$$

Separating real/imaginary parts and solving the 2×2 system for the unknown ω_3, ω_4 :

$$\omega_3 = \frac{a\omega_2}{b} \frac{\sin(\theta_4 - \theta_2)}{\sin(\theta_3 - \theta_4)}, \quad \omega_4 = \frac{a\omega_2}{c} \frac{\sin(\theta_2 - \theta_3)}{\sin(\theta_4 - \theta_3)}.$$

A negative result means the link rotates clockwise (opposite to the assumed CCW positive).

Velocity of Points

Once the angular velocities are known, the velocity of any point follows from $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, i.e. for a point at link angle θ and radius R from the pivot, $\mathbf{v} = R\omega(-\sin\theta, \cos\theta)$.

Pin A (end of crank):

$$\mathbf{V}_A = a\omega_2(-\sin\theta_2, \cos\theta_2), \quad |\mathbf{V}_A| = a\omega_2.$$

Pin B (relative-velocity / difference equation):

$$\mathbf{V}_B = \mathbf{V}_A + \mathbf{V}_{BA}, \quad \mathbf{V}_{BA} = b\omega_3(-\sin\theta_3, \cos\theta_3).$$

Coupler point P at distance R_{PA} from A, oriented at $(\theta_3 + \delta_3)$:

$$\mathbf{V}_P = \mathbf{V}_A + R_{PA}\omega_3(-\sin(\theta_3 + \delta_3), \cos(\theta_3 + \delta_3)).$$

Exam focus. $\mathbf{V}_B$ must come out the same whether computed from link 4 ($c\omega_4$, $\perp$ to O_4B) or from $\mathbf{V}_A + \mathbf{V}_{BA}$. Use this as your exam check.

Slider-Crank Velocities

For the offset slider-crank, differentiating the loop gives the coupler angular velocity and the slider (linear) velocity:

$$\omega_3 = -\frac{a\omega_2 \cos\theta_2}{b \cos\theta_3}, \quad \dot{d} = -a\omega_2 \sin\theta_2 - b\omega_3 \sin\theta_3.$$

The **velocity of slip** at a sliding joint is the relative sliding velocity along the slider axis; for the simple crank-slider it equals $\dot{d}$.

Inverted slider-crank: the slip occurs along the rotating link, so the velocity difference equation includes a slip term: $\mathbf{V}_B = \mathbf{V}_{B'} + \mathbf{V}_{\text{slip}}$, where B and B' are coincident points on the two links at that instant.

Instant Centres of Velocity

An **instant centre (IC)** is a point common to two links having the *same instantaneous velocity*; the relative motion of the two links is pure rotation about it.

- Number of ICs: $N = \frac{n(n-1)}{2}$ for n links.
- Pin joints are obvious ICs ($I_{1,2}, I_{2,3}, I_{3,4}, I_{1,4}$).
- **Kennedy's theorem:** for any three links, their three ICs lie on one straight line. Use it to locate the hidden ICs $I_{1,3}$ and $I_{2,4}$.

Exam focus. $I_{2,4}$ lies on the line through $I_{1,2}I_{1,4}$ and the line through $I_{2,3}I_{3,4}$; $I_{1,3}$ lies on $I_{1,2}I_{2,3}$ and $I_{1,4}I_{3,4}$. The angular-velocity ratio follows from the distances of $I_{2,4}$ to the two ground pivots.

Tutorial Solutions (Tutorial 5 — Velocity)

P6-5 — Four-Bar Angular Velocities and Coupler-Point Velocity (Row a)

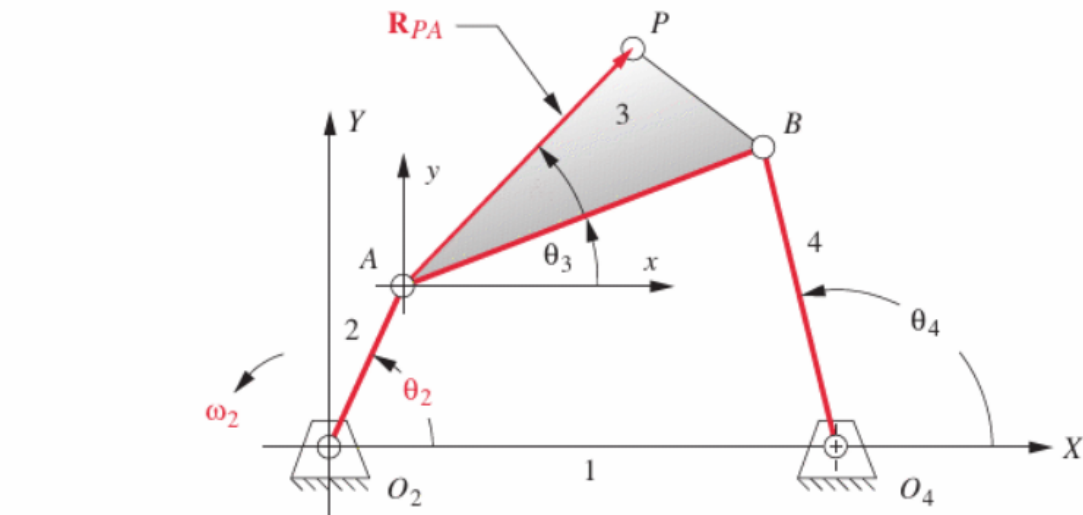


FIGURE P6-1

Figure P6-1: general four-bar with coupler point P at $\mathbf{R}_{PA}$, angle δ_3 .

Given:

Row a (Table P6-1): $a = L_2 = 2$, $b = L_3 = 7$, $c = L_4 = 9$, $d = L_1 = 6$, $\theta_2 = 30^\circ$, $\omega_2 = 10$ rad/s, $R_{PA} = 6$, $\delta_3 = 30^\circ$. From position analysis (P4-7, open): $\theta_3 = 88.8^\circ$, $\theta_4 = 117.3^\circ$.

Find:

ω_3 , ω_4 , and the velocity of point P (open configuration).

Solution:

Angular velocities:

$$\omega_3 = \frac{(2)(10)}{7} \cdot \frac{\sin(117.3^\circ - 30^\circ)}{\sin(88.8^\circ - 117.3^\circ)} = 2.857 \cdot \frac{0.999}{-0.477} = \boxed{-5.99 \text{ rad/s}},$$

$$\omega_4 = \frac{(2)(10)}{9} \cdot \frac{\sin(30^\circ - 88.8^\circ)}{\sin(117.3^\circ - 88.8^\circ)} = 2.222 \cdot \frac{-0.856}{0.477} = \boxed{-3.99 \text{ rad/s}}.$$

Both negative $\Rightarrow$ links 3 and 4 rotate clockwise.

Velocity of A:

$$\mathbf{V}_A = (2)(10)(-\sin 30^\circ, \cos 30^\circ) = (-10.0, 17.32) \text{ in/s}, \quad |\mathbf{V}_A| = 20 \text{ in/s}.$$

Velocity of coupler point P (orientation $\theta_3 + \delta_3 = 118.8^\circ$):

$$\mathbf{V}_{PA} = (6)(-5.99)(-\sin 118.8^\circ, \cos 118.8^\circ) = (31.5, 17.4) \text{ in/s},$$

$$\mathbf{V}_P = \mathbf{V}_A + \mathbf{V}_{PA} = (21.5, 34.7) \text{ in/s}, \quad \boxed{|\mathbf{V}_P| = 40.8 \text{ in/s at } 58.2^\circ}.$$

Check on $\mathbf{V}_B$: from link 4, $\mathbf{V}_B = (9)(-3.99)(-\sin 117.3^\circ, \cos 117.3^\circ) = (31.9, 16.5)$; from $\mathbf{V}_A + \mathbf{V}_{BA}$ with $\mathbf{V}_{BA} = (7)(-5.99)(-\sin 88.8^\circ, \cos 88.8^\circ) = (41.9, -0.85)$, giving $(31.9, 16.5)$ ✓

P6-7 — Offset Crank-Slider Velocities and Slip (Row a)**Given:**

Row a (Table P7-2 / P6-2): crank $a = L_2 = 1.4$, coupler $b = L_3 = 4$, offset $c = 1$, $\theta_2 = 45^\circ$, $\omega_2 = 10$ rad/s. From position analysis (open): $\theta_3 = 0.14^\circ$, slider $d = 4.99$.

Find:

angular velocity ω_3 , the velocities of pins A and B, and the velocity of slip at the sliding joint.

Solution:

$$\omega_3 = -\frac{a\omega_2 \cos \theta_2}{b \cos \theta_3} = -\frac{(1.4)(10) \cos 45^\circ}{4 \cos 0.14^\circ} = \boxed{-2.48 \text{ rad/s}}.$$

$$|\mathbf{V}_A| = a\omega_2 = (1.4)(10) = 14 \text{ in/s } (\perp \text{ to crank}).$$

$$\dot{d} = -a\omega_2 \sin \theta_2 - b\omega_3 \sin \theta_3 = -(1.4)(10) \sin 45^\circ - (4)(-2.48) \sin 0.14^\circ = \boxed{-9.88 \text{ in/s}}.$$

Since the slider B translates only, $\mathbf{V}_B = \dot{d}$ along the slide and the **velocity of slip** at the sliding joint is $\boxed{9.88 \text{ in/s}}$ (directed in $-X$).